Tshwane University of Technology

We empower people

Faculty of Information and Communication Technology

Department of Computer Science

STUDY GUIDE

NQF LEVEL	NQF CREDITS	MODULE NAME	MODULE CODE	YEAR	SEMESTER
7	15	Integrated Software Project	ISJ107V	2026	1 & 2
QUALIFICATION			SAQA ID:		
Advanced Diploma in Computer Science					

Compiled by Mr. HD Masethe
Revised on: 30 January 2026

Contents

SECTION A – INTRODUCTION

1. Welcome

Welcome to **Integrated Software Project**

Integrated Software Project is a project-based subject, with each student assigned to a group. The students are required to show competence in the utilization of newer ICT technologies and trends as part of the research design and development of a computer-based system.

As part of the technical designs and development of the systems the student should show proficiency and competence in the following concepts:

A. Investigate latest developments and trends in ICT systems development
B. Applying Object Oriented Analysis and Design principles.
C. Applying UI design and UX
D. Incorporating new technologies and trends into system design and development.

2. Introduction

This Integrated Software Project module is a 15-credit module with core learning modules on 7 which presents students with the opportunity to apply and extend their practical knowledge acquired in other modules completed prior to this one by completing an industry-related Information Technology software project similar to projects found in a workplace environment, incorporating relevant current technologies. This module will not only enable students to apply but also appreciate the usefulness of their skills and knowledge acquired thus far in this qualification. The final product of the project should be a workable system illustrating the application of current trends. The technical knowledge obtained in this module together with the communication skills, the skills to work in a team and presentation skills will prepare students for the workplace. Graduates should have the ability to make effective presentations to a range of audiences about technical problems and their solutions. This may involve face-to-face, written, or electronic communication. They should be prepared to work effectively as members of teams. Students should be able to manage their own learning and development, including managing time, priorities, and progress.

3. Name of the module

Integrated Software Project- Advanced Diploma in Computer Science

4. Module Credits/Weight

- 15 credits.

5. Custodian department of this module

The custodian department of this module is the Department of Computer Science.

6. Purpose of the module

After completion of this course, students should have the ability to analyses and design software solutions to industry-related Information Technology problems and utilize the required technical skills to effectively implement the designed solutions in distributed IT environments. The students should also demonstrate the ability to research a new information technology and present an analysis thereof.

This module forms part of the qualification Advanced Diploma in Computer Science and will individually contribute to achieving the exit level outcome of:

- ELO 4: Apply problem solving skills and the knowledge of computer science to solve real world problems.

This module is placed in the fourth year as it helps to prepare student to use different trends in software development methodologies, networking principles, project management, database design and implementation and programming.

The NQF of this module is 7 and on this level, it is expected that learning should display the following learning achievements:

a. Scope of knowledge, in respect of which a learner is able to demonstrate: detailed knowledge of the main software development methodologies, networking principles, project management, database design and implementation and programming, including an understanding of and the ability to apply the key terms, concepts, facts, principles, rules and theories of that field, discipline or practice to unfamiliar but relevant contexts; and knowledge of an area or areas of specialisation and how that knowledge relates to other fields, disciplines or practices.
b. Knowledge literacy, in respect of which a learner is able to demonstrate an understanding of different forms of knowledge, schools of thought and forms of explanation within software development methodologies, networking principles, project management, database design and implementation and programming, in terms of operation or practice, and awareness of knowledge production processes.

c. Method and procedure, in respect of which a learner is able to demonstrate the ability to evaluate, select and apply appropriate methods, procedures or techniques in investigation or application processes within a defined context.

d. Problem solving, in respect of which a learner can demonstrate the ability to identify, analyse and solve problems in unfamiliar contexts, gathering evidence and applying solutions based on evidence and procedures appropriate to software development methodologies, networking principles, project management, database design and implementation and programming.

e. Ethics and professional practice, in respect of which a learner is able to demonstrate an understanding of the ethical implications of decisions and actions within an organisational or professional context, based on an awareness of the complexity of ethical dilemmas.

f. Accessing, processing and managing information, in respect of which a learner is able to demonstrate the ability to evaluate different sources of information, to select information appropriate to the task, and to apply well-developed processes of analysis, synthesis and evaluation to that information.

g. Producing and communicating information, in respect of which a learner is able to demonstrate the ability to present and communicate complex information reliably and coherently using appropriate academic and professional or occupational conventions, formats and technologies for a given context.

h. Context and systems, in respect of which a learner is able to demonstrate the ability to make decisions and act appropriately in familiar and new contexts, demonstrating an understanding of the relationships between systems, and of how actions, ideas or developments in one system impact on other systems.

i. Management of learning, in respect of which a learner is able to demonstrate the ability to evaluate performance against given criteria, and accurately identify and address his or her task-specific learning needs in a given context, and to provide support to the learning needs of others where appropriate.

j. Accountability, in respect of which a learner is able to demonstrate the ability to work effectively in a team or group, and to take responsibility for his or her decisions and actions and the decisions and actions of others within well-defined contexts, including the responsibility for the use of resources where appropriate.

The graduate attributes of the institution expect students to be well rounded individuals which will demonstrate knowledge, skills and applied competencies. By employing the proper outcomes, assessment methods and level descriptors this module will contribute to the knowledge and skills involved in software development methodologies, networking principles, project management, database design and implementation and programming and applying these skills to either the working environment or their own lives.

The students will be employed because of an array of skills and knowledge they will acquire. Industry expects students to:

- Communicate effectively,
- Be critical thinkers,

- Be team players and yet independent thinkers.

Through this module students will acquire skills and applied competencies by doing investigation and applying solutions to provided issues through group work and presentation of solutions. The intention of the module is to ensure that the student is well-rounded, and this can only be ensured if the individuals can interact effectively with their changing environment and act responsive and responsible within a variety of social and cultural settings.

The students will be employable because of an array of skills and knowledge they will acquire. Industry expects students to: communicate effectively, to be critical thinkers, team players and yet independent thinkers. Through this module, student will acquire skills and applied competencies by doing investigation and applying solutions to provided issues through group work and presentation of solutions.

This module forms part of the following qualifications:

- Advanced Diploma in Computer Science.
- Advanced Diploma in Informatics.
- Advanced Diploma in Information Technology.

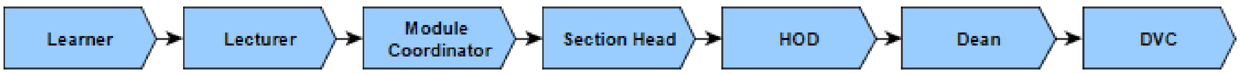
For the Advanced Diploma in Computer Science this module will individually contribute to achieving the exit level outcome of:

- ELO1: Demonstrate understanding of Software Engineering concepts.
- ELO 2: Evaluate and develop strategies for effectively managing software development.
- ELO 3: Apply techniques for Software Requirements.
- ELO 4: Design software solution including architecture and environment.
- ELO 5: Develop and administer a software quality assurance plan according to applicable standards including Software Engineering Code of Ethics.
- ELO 6: Develop software, administer and configuration plan according to applicable standards

The intent of the Integrated Software Project module is to ensure that the students are well-rounded, and this can only be ensured if the individuals can interact effectively with their changing environment and act responsive and responsible within a variety of social and cultural settings. Aligned to the graduate attributes and ELO's, students will be guided through activities like problem-solving and reflexive practices. In addition, students will acquire the following Knowledge: understand key terms, concepts, facts, principles, rules and theories. Skills: Select and apply standard methods, procedures and/or techniques Plan, manage and implement processes within a supported environment. The module will require students to work individually and in groups thereby contributing towards both personal and professional ethics. As the students' progress in their study, they will be required to present their work to other fellow students, thereby developing their presentation and communication skills (skills identified as graduate attributes by the university). The technical knowledge obtained in this module together with the communication skills and presentation skill will prepare the students for the workplace. This tie in with the instructional goal - To prepare diverse students for rewarding careers and responsible citizenry by providing a student-centred learning experience that is underpinned by a scholarship of teaching and learning.

7. Code of conduct

- Students may not be late for class. (Do not enter the class if you are late)
- Students should attend all classes. (The 85% class attendance rule applies)
- Cell phones should be switched off during class times.
- No eating or drinking in the classes.
- Students should strictly follow the schedule, unless otherwise stated. (page 4 of this study guide)
- Students should also look on the notice board outside the lecturer's office for updates on marks and communication.
- Each student should have a textbook.
- Students should prepare for each class by researching the module.
- Students can see lecturer during consultation times, otherwise make an appointment with the lecturer for another time.
- Students should submit their sick note or death certificate within the specified time period.
- Since 2011 a student is not allowed to register more than three times for a module.
- All issues pertaining to the module must be reported to your lecturer. If the issue cannot be resolved adequately it must be escalated to the module coordinator. If the issue is still not resolved, it must be escalated to the section head. If the issue is still not resolved, it can be escalated to the HOD of the module's custodian department (see section A). Students may only report issues to the Dean of the faculty if the issue cannot be resolved by the HOD. Under no circumstances will issues be addressed if the proper reporting chain was not followed (see the figure below)



SECTION B – ORGANISATIONAL COMPONENT

8. Contact information of custodian department of this module

The custodian department of this module is the department of Computer Science.

Name & Surname	Campus	Office Location	Contact number and E-mail	Role in Programme
Me Sinnah Mokhutso	Soshanguve South	20-203	012 3829631 MokhutsoMS@tut.ac.za	Administrator
Me Angelina Shonaphi Bhembe	E-Malahleni	14-G37	013 653-3165 BhembeAS@tut.ac.za	Administrator
Me Maesela Lebelo	Polokwane	1-G247	015 287-0757 LebeloM@tut.ac.za	Administrator

9. Contact information of lecturers

This module has one primary **module coordinator** that is responsible for coordinating the activities of the module. If this module is offered at one of the other learning sites and the primary module coordinator does not reside on the leaning site, the activities of this module is managed by the primary module coordinator via a **site coordinato**r. There is one **facilitator** assigned to facilitate the teaching of each group.

9.1. Module coordinator/s

Module Coordinator	Dr. HD Masethe (Subject Head)
Office Location	20 – 102 (Soshanguve South Campus)
Office Telephone Number	012 382 9714
Email address	masethehd@tut.ac.za
Module Coordinator	Dr. K Sigama
Office Location	Building 3-G03 (Polokwane)
Office Telephone Number	012 382 0961
Email address	sigamak@tut.ac.za

Module Coordinator	Dr. B Buitendag
Office Location	20-212 (Soshanguve South Campus)
Office Telephone Number	012 382 9631
Email address	buitendagaak@tut.ac.za

Module Coordinator	Mr. S Marebane
Office Location	Room 14-G35 (eMalahleni)
Office Telephone Number	013 653-3136/3165
Email address	marebanesm@tut.ac.za

10. Library Contact Details

Soshanguve Faculty Librarian	Ms Rachel Raisibe Ntsoane
Office Location	Ground floor in library
Office Telephone Number	012 799 9509
Email address	NtsoaneRR@tut.ac.za

11. Student Support Contact Details

Student Support Services	Dr Shafeeka Dockrat
Office Location	Pta: Building 5, Room 5-705
Office Telephone Number	+27 12 382 4260
Email address	DockratS@tut.ac.za

12. Time table

The timetable for class attendance is obtainable from the leaner management system (LMS). Classes are conducted every Saturday 11h30-14h30 using MS Teams.

Microsoft Teams meeting

Join: https://teams.microsoft.com/meet/32658788875868?p=aU45iRqwP6swtHJJkS
Meeting ID: 326 587 888 758 68
Passcode: 62g4pE2E

13. Consultation

Consultation time slots will be displayed on each lecturer's office door. In the case where the time slots are not displayed students must please contact the lecturer one day in advance via email and arrange a consultation session. Please remember to mention your name & surname, student number, module code and topic to be discussed during the consultation session.

14. Schedule

The schedule indicates all-important dates for activities such as, class activities, assignment due dates, class tests, excursions, practical's, project due dates, computer-based tests submission dates etc. Please ensure that you follow the schedule of your assigned group.

TUT Schedule	Learning Outcomes	Assessment Criteria	Activity /Presenter
Week 1 2-7 Feb	Introduction Unit 1: **Outcomes:** A.1. Demonstrate an understanding of newer developments and trends in ICT tool and service development.	**Introduction of the study Guide** • The literature is surveyed to identify and describe evolving trends and technologies. • Various modern day software development trends, technologies and architectures are outlined against their use and applicability.	Dr. HD Masethe Dr. K. Sigama Mr. S. Marebane Ms L Baloyi
Week 2 9-14 Feb	**Outcomes:** A.1. Demonstrate an understanding of newer developments and trends in ICT tool and service development.	• The appropriateness of a particular technology is assessed against a set of design specifications. • A particular technology or combination of technologies towards the development and design of a solution are recommended.	Dr. HD Masethe Dr. K. Sigama Mr. S. Marebane Ms L Baloyi
Week 3 23-28 Feb	**Outcomes:** B.1. Demonstrate proficiency in the use of Object Oriented Analysis and Design principles using UML.	• A given scenario of business case towards the development of user and systems requirements is analysed. • Various user requirements models are created based on a particular scenario.	Mr. Marebane **Assignment 1 (10%)**
Week 4 2-7 Mar	Outcomes: C.1. Demonstrate an understanding of modern trends in UI design and UX		

Week 5 9-14 Mar	**Feedback**		
Week 6 16-21 Mar			
Week 7 23-28 Mar	**Test Week 1** Outcomes: D.1. Incorporate newer technologies and trends as part of the as part of the system's design and development	The concept of DevOps is demonstrated in terms of its theoretical principles and practical implementation. • Cloud services and technologies are included as part of the designs and implementation of the solution. • Various relevant technologies are incorporated as part of the development of the software solution. • D.1.4 - Best practices are outlined as part of the development of the solution. • D.1.5 - Multi-platform functionality and interoperability is incorporated as part of the solutions functional design based	Dr B Buitendag

		on the requirements.	
Week 8 30 Mar - 4 Apr		Recess	
6-11 Apr			
Week 9 13-18 Apr	**Outcomes:** C.1. Demonstrate an understanding of modern trends in UI design and UX.	• The conceptual design of an applicable UI based on the design specifications incorporating UX principles is presented. • The concept of UX design is discussed. Different designs of UIs designed according to UX principles for different platforms are presented.	Dr. TJ Ramabu Dr. K. Sigama Mr. S Marebane
Week 10 20-25 Apr	**Outcomes:** D.1. Incorporate newer technologies and trends as part of the as part of the system's design and development.	• The concept of DevOps is demonstrated in terms of its theoretical principles and practical implementation. • Cloud services and technologies are included as part of the designs and implementation of the solution. • Various relevant technologies are incorporated as part of the development of the software solution. • D.1.4 - Best practices are outlined as part of the development of the solution. • D.1.5 - Multi-platform functionality and interoperability is incorporated as part of the	Mr. HD Masethe Dr. K. Sigama Dr. TJ Ramabu

		solutions functional design based on the requirements.	
Week 11 27 Apr - 2 May	**Test Week 2**	**Feedback/Presentations on**	**Assignment 2 (10%)**
Week 12 4-9 May	**Test Week 2**	**Feedback**	
Week 13 11-16 May		**Feedback**	
Week 13 18-23 May		**Feedback**	
Week 14 25-30 May		**Feedback**	**Assignment 3 (10%)**

Week 15 1-6 Jun			
8-13 Jun			
July		**Student Presentation on prototype**	**Assignment 4 (20%)**
Septembe r		**Student Presentation on final Project**	**Project (50%)** The actual system working or simulated in a real live environment, including having its components hosted on different machines.

SECTION C – PRESCRIBED AND RECOMMENDED RESOURCES

15. Prescribed Textbook

This book is essential for the successful completion of this course. You are strongly advised to acquire the resource.

BOOKS:

Erl, T. 2017. Service-Oriented Architecture: Analysis and Design for Services and Microservices. 2nd Ed. Prentice Hall.

Hall, G. 2017. Adaptive Code: Agile coding with design patterns and SOLID principles. 2nd Ed. Microsoft Press.

Schach, R. 2011. Object-Oriented and Classical Software Engineering. 8th Ed. New York: McGraw-Hill.

16. Additional Module Resource Information

- Student could be given handouts from other relevant books.
- Computers

17. Recommended Additional Resources

- https://www.tutorialspoint.com/software_engineering/software_engineering_overview.htm
- https://www.javatpoint.com/software-engineering-tutorial
- https://learn.saylor.org/course/CS302
- https://www.jcse.org.za/
- https://www.springer.com/journal/766
- https://www.computer.org/education/bodies-of-knowledge/software-engineering

18. Recommended Electronic resources

http://www.SoftwareEngineering-9.com

www.pearsonglobaleditions.com/Sommerville

www.youtube.com/user/SoftwareEngBook

19. Supplied Resources

20. E-Learning platform

This module will make use of Mytutor for its formal e-learning platform. Last minute important notifications, like changes in lecture venue, time, deadlines for assignments, etc. will be placed on this platform. Electronic resources will also be distributed via this platform.

SECTION D – MODULE DESCRIPTION

21. Alignment of this module with the relevant exit level outcomes / NQF 7

The students should be able to do the following after the completion of the module:

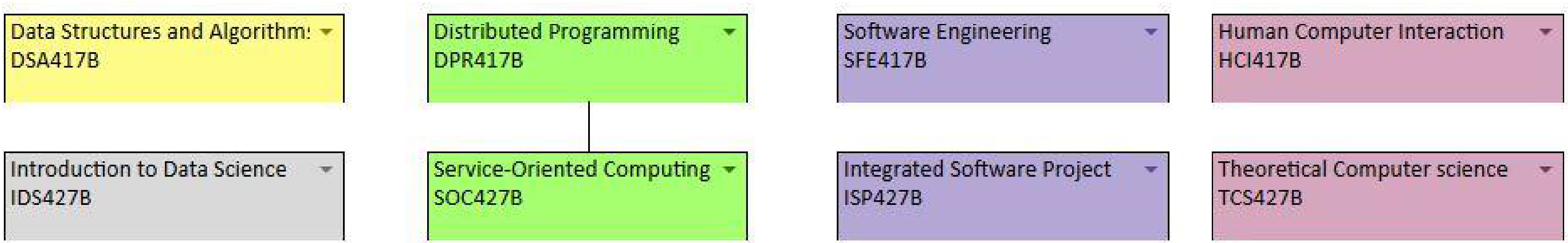
- Scope of knowledge, in respect of which a learner is able to demonstrate: detailed knowledge of the main areas of one or more fields, disciplines or practices, including an understanding of and an ability to apply the key terms, concepts, facts, principles, rules and theories of that field, discipline or practice to unfamiliar but relevant contexts; and knowledge of an area or areas of specialisation and how that knowledge relates to other fields, disciplines or practices
- Knowledge literacy, in respect of which a learner is able to demonstrate a understanding of different forms of knowledge, schools of thought and forms of explanation within an area of study, operation or practice, and an awareness of knowledge production processes
- Method and procedure, in respect of which a learner is able to demonstrate an ability to evaluate, select and apply appropriate methods, procedures or techniques in processes of investigation or application within a defined context
- Problem solving, in respect of which a learner is able to demonstrate an ability to identify, analyse and solve problems in unfamiliar contexts, gathering evidence and applying solutions based on evidence and procedures appropriate to the field, discipline or practice
- Ethics and professional practice, in respect of which a learner is able to demonstrate an understanding of the ethical implications of decisions and actions, within an organisational or professional context, based on an awareness of the complexity of ethical dilemmas f. Accessing, processing and managing information, in respect of which a learner is able to demonstrate an ability to evaluate different sources of information, to select information appropriate to the task, and to apply well-developed processes of analysis, synthesis and evaluation to that information
- Producing and communicating information, in respect of which a learner is able to demonstrate an ability to present and communicate complex information reliably and coherently using appropriate academic and professional or occupational conventions, formats and technologies for a given context
- Context and systems, in respect of which a learner is able to demonstrate an ability to make decisions and act appropriately in familiar and new contexts, demonstrating an understanding of the relationships between systems, and of how actions, ideas or developments in one system impact on other systems

Management of learning, in respect of which a learner is able to demonstrate an ability to evaluate performance against given criteria, and accurately identify and address his or her task-specific learning needs in a given context, and to provide support to the learning needs of others where appropriate

- Accountability, in respect of which a learner is able to demonstrate an ability to work effectively in a team or group, and to take responsibility for his or her decisions and actions and the decisions and actions of others within well-defined contexts, including the responsibility for the use of resources where appropriate.

22. Advanced Diploma in Computer Science

Adv. Diploma in Computer Science (ADCS12?)



The outline of the module is as follows:

A. Professional Practice.

B. Software Process.

C. Software modelling, analysis and specification.

D. Software Design.

E. Software Quality, Verification and Validation.

F. Software Configuration management.

23. Learning outcomes, assessment criteria, teaching activities, and Assessment method

Each **learning goal** has a set of **learning outcomes** that will be assessed using **assessment criteria**. The tables below show the learning outcomes, assessment criteria, teaching activities, and assessment methods.

UNIT 1		
Learning outcome: Outcomes: A.1. Demonstrate an understanding of newer developments and trends in ICT tool and service development.		
Assessment Criteria *(noun + verb + measurable standard)*	**Teaching activities**	**Assessment method**
A.1.1 - A.1.1 The literature is surveyed to identify and describe evolving trends and technologies. A.1.2 - Various modern day software development trends, technologies and architectures are outlined against their use and applicability. A.1.3 - The appropriateness of a particular technology is assessed against a set of design specifications. A particular technology or combination of technologies towards the development and design of a solution are recommended.	**Lecturer controlled:** Explain the concepts, demonstrate the application of the concepts using problem based or case studies. **Peer controlled**: Reinforce the concepts in practical and tutorial sessions. **Student controlled:** demonstrate the understanding by tests, assignments, presentations.	**Formative: written assignments** **Summative: End of semester project**

UNIT 2		
Learning outcome: B.1. Demonstrate proficiency in the use of Object Oriented Analysis and Design principles using UML.		
Assessment Criteria *(noun + verb + measurable standard)*	**Teaching activities**	**Assessment method**
B.1.1 - A given scenario of business case towards the development of user and systems requirements is analysed.	**Lecturer controlled:** Explain the concepts, demonstrate the	**Formative:** written assignments

B.1.2 - Various user requirements models are created based on a particular scenario. B.1.3 - The functional requirements of a system are modelled. B.1.4 - Various design patterns are recognised, outlined and contrasted. An applicable design pattern or patterns are used as part of the functional designs of the system.	application of the concepts using problem based or case studies. **Peer controlled**: Reinforce the concepts in practical and tutorial sessions. **Student controlled:** demonstrate the understanding by tests, assignments, presentations.	**Summative**: End of semester project

UNIT 3 **Learning outcome: C.1. Demonstrate an understanding of modern trends in UI design and UX.**		
Assessment Criteria *(noun + verb + measurable standard)*	**Teaching activities**	**Assessment method**
C.1.1 - The conceptual design of an applicable UI based on the design specifications incorporating UX principles is presented. C.1.2 - The concept of UX design is discussed. C.1.3 - Different designs of UIs designed according to UX principles for different platforms are presented.	**Lecturer controlled:** Explain the concepts, demonstrate the application of the concepts using problem based or case studies. **Peer controlled**: Reinforce the concepts in practical and tutorial sessions.	**Formative:** written assignments, tests, quizzes, **Summative**: End of semester project

	Student controlled: demonstrate the understanding by tests, assignments, presentations.	

UNIT 4 **Learning outcome:** D.1. Incorporate newer technologies and trends as part of the as part of the system's design and development.		
Assessment Criteria *(noun + verb + measurable standard)*	**Teaching activities**	**Assessment method**
D.1.1 - The concept of DevOps is demonstrated in terms of its theoretical principles and practical implementation. D.1.2 - Cloud services and technologies are included as part of the designs and implementation of the solution. D.1.3 - Various relevant technologies are incorporated as part of the development of the software solution. D.1.4 - Best practices are outlined as part of the development of the solution. D.1.5 - Multi-platform functionality and interoperability is incorporated as part of the solutions functional design based on the requirements.	**Lecturer controlled:** Explain the concepts, demonstrate the application of the concepts using problem based or case studies. **Peer controlled**: Reinforce the concepts in practical and tutorial sessions. **Student controlled:** demonstrate the understanding by tests, assignments, presentations.	**Formative:** written assignments, tests, quizzes, **Summative**: End of semester project

As presented in the tables above, each learning outcome will be assessed with an assessment method. Below is a table that describes what these assessment methods entail.

Assessment Method	Example/description	Possible uses/assessment tasks	Where to use
Alternative response questions	True/False; Yes/No questions; multiple choice	• Recall of information • Ability to discriminate	• Formative • Summative
Assertion/reason questions	Consists of an assertion and supporting explanation. The learner has to decide whether the assertion and explanation are true, and if true, whether the explanation is a valid reason for the assertion. Sometimes the learner is asked to select his/her answer from a list of possibilities, e.g. True; True + Valid; True + Invalid.	• Ability to weigh up options and to discriminate	• Formative • Summative • RPL
Aural/oral tests	These are mainly used to generate evidence on learners' ability to listen, interpret, communicate ideas and sustain a conversation in the language of assessment.	• Interpretation of ideas • Expression of ideas	• Formative • Summative • RPL
Completion questions/short answer questions	Learners are presented with a question with a pre-determined answer consisting of a few words, or may be given a statement where key words are omitted. They are then required to complete the statement by filling in the word(s). Such questions may also involve the use of numbers, diagrams and graphs.	• Recall of factual information • Test understanding and application of knowledge, e.g. in mathematical concepts	• Formative • Summative • RPL
Examinations / tests	These usually consist of a range of questions. Learners are required to respond to questions within a specified time.	• Recall of information • Cognitive skills such as problem solving or analyses	• Formative • Summative • RPL
Extended response questions	These are usually in a written form. There are few restrictions on the content and form of the response. Continuous prose is normally required, but there may be limits on the length and/or time allocated.	• Open-ended debates or other responses • Arguments • Reports	• Formative • Summative • RPL
Grid questions /matching questions	Grid questions and matching questions are variants of each other. The learner is presented with two lists – a set of statements and a set of responses. The learner is required to indicate which response from the second list corresponds or matches each statement in the first list. Grid questions are presented in grid format. They differ from the other selected-response assessment instruments	• Recall of information • Application of knowledge	• Formative • Summative • RPL

	in that each question may have more than one correct response and each response may be used more than once.		
Multiple choice questions	Multiple choice questions consist of an incomplete statement or a question, followed by plausible alternative responses from which the learner has to select the correct one. Outcomes involving higher order analytical skills are probably more validly assessed by means of free-response assessment instruments such as extended response questions, but multiple-choice questions can be useful if carefully constructed.	• Recall of information • Check understanding; analyses	• Formative • Summative • RPL
Oral questions / restricted response questions	The form and content of the response is limited by the way in which the question is asked. These questions do not have pre-determined correct answers (as in short answer questions, etc.) and the assessor has to exercise his/her professional judgment when interpreting learner's responses.	• Allows for self-expression when questions are oral Supports observation of tasks where underpinning knowledge and • understanding are tested	• Formative • Summative (small groups only) • RPL
Personal interviews	A personal interview is probably the oldest and best-known means of eliciting information directly from learners. It combines two assessment methods, namely observation and questioning. An interview is a dialogue between the assessor and the learner, creating opportunities for learner questions.	• A range of applications using different forms of questions, particularly • Open-ended questions • Guidance and support to the learner	• Formative • Summative (small groups only) • RPL
Questionnaires	A questionnaire is a structured written interview consisting of a set of questions relating to particular areas of performance. Unlike a personal interview, it is administered and judged under standard conditions.	• Assessment of outcomes particularly concerned with attitudes, feelings, interests and experiences	• Formative • Summative • RPL
Structured questions	A structured question consists of a stem (which describes a situation), followed by a series of related questions. The stem can be text, a diagram, a picture, a video, etc.	• Recall of information • Application of knowledge and understanding • Analyses, Debates, Arguments	• Formative • Summative • RPL
Assignments	A problem-solving exercise with clear guidelines and a specified length. More structured and less open-ended than projects, but they do not necessarily involve strict adherence to a prescribed procedure, and they are not concerned exclusively with manual skills.	• Problem-solving around a particular topic	• Formative • Summative • RPL
Case studies	A description of an event concerning a real-life or simulated situation, usually in the form of a paragraph or text, a video, a picture or a role-play exercise. This is	• Analyses of situations • Drawing conclusions	• Formative • Summative

	followed by a series of instructions to elicit responses from learners. Individuals or small groups may undertake case studies.	• Reports on possible courses of action	• RPL
Logbooks	A useful means of assessing learner's progress and achievements. It should have clear instructions for use and give guidance on how essential information is to be recorded.	• In a workplace – monitor and check activities; record processes; record of achievements	• Formative • Summative • RPL • (Learner has to be in workplace)
Practical exercises / demonstrations	An activity that allows learners to demonstrate manual and/or behavioural skills. The assessment may be based on the end-result of the activity (the product), or the carrying-out of the activity (the process), or a combination of both	• Demonstration of skill	• Formative • Summative • RPL (Not always practical – logistics)
Portfolios	A collection of different types of evidence relating to the work being assessed. It can include a variety of work samples. Portfolios are suitable for long-term activities. It is important that the evidence in the portfolio meet the requirements of sufficiency and currency. The learner and assessor usually plan the portfolio jointly as sources of evidence may vary. The learner is then responsible for the collection of evidence and the compilation of the portfolio.	• Recognition of prior learning and experience • Assessment of long-term activities related to each other • Assessment where direct observation may be difficult	• Summative • RPL • (Not recommended for formative)
Projects	A project is any exercise or investigation in which the time constraints are more relaxed. Projects are: • Practical • Comprehensive and open-ended • Tackled without close supervision, but with assessor guidance and support Projects can involve individuals or a group of learners. The choice of the project is directed by the assessor, usually by providing the learner with a topic or brief for the investigation	• Comprehensive range of skills can be assessed • Integration of activities within and across unit standards or different parts of a qualification	• Summative • RPL • (Not recommended for formative)
Role-plays	Learners are presented with a situation, often a problem or an incident, to which they have to respond by assuming a particular role. The enactment may be unrehearsed, or the learner may be briefed in the particular role to be played. Such assessments are open-ended and are person centred.	• Assessment of a wide range of behavioural and interpersonal skills	• Formative • Summative • RPL
Reflective journal	A reflective journal gives learners the opportunity to critically reflect on their own learning, to express their thoughts and experiences and to present this in an acceptable way. Even though it is a form of self-assessment, it can be submitted for assessment.	• It gives the assessor a unique opportunity to follow the thought-processes of a learner and to monitor the	• Formative

		way a learner thinks and grows • Critical evaluation of progress by the learner	
Self-assessment	A checklist, questionnaire completed by a learner, notes jotted down, or other forms of structured self-assessment undertaken after an action, demonstration, oral examination, etc.	• Critical evaluation of progress by learner	• Formative
Peer assessment	Assessment by the learner's peers, usually in the form of a checklist.	• Assessment of paired or group activities • Assessment of teamwork	• Mainly formative • Summative if team-, group work part of outcomes

SECTION E – ASSESSMENT

24. Assessment Schedule

The learner will write formal semester tests during the course of this module. Learners, who did not write one of the formal semester tests due to a reason other than sickness or extraordinary circumstances, will forfeit the mark for predicate. Each formal assessment will count a percentage towards the student's predicate mark as shown in the table below.

The examiner reserves the right to change the above structure as needed. Predicate marks are put on the faculty notice boards. If you have queries about your mark, you must immediately consult your lecturer before predicate day. Once the predicate mark is entered on TUT's student management system, the mark cannot be changed.

The predicate mark and the examination mark each contribute 50% towards the final mark as shown in the table below.

ASSESSMENT PLAN										
					PURPOSE					
ASSESSMENT TASK	**INSTRUMENT** (Assignment, test, worksheet etc.)	**DESCRIPTION** (Short description)	**METHOD** (Observation, product or questioning)	**TOOL** (Rubric, memo, checklist etc.)	**DIAGNOSTIC (D)**	**FORMAL (M) INFORMAL (I)**	**FORMATIVE (F) SUMMATIVE (S)**	**PROMOTION (P)**	**WEIGHT (%)**	**DUE DATE (Optional)**
1	Assignment 1	Project Proposal	Question	Rubric	D	M	F	P	10	
2	Assignment 2	The formal system specification document i.e. technical hardware requirements document, incorporating UML designs. (formal Proposal of the system)	Question	Rubric	D	M	F	P	10	
3	Assignment 3	The project documentation, incorporating UML designs.	Question	Rubric	D	M	F	P	15	
4	Assignment 4	Document outlining the use of applicable technologies.	Question	Rubric	D	M	F	P	15	
5	Project	The actual system working or simulated in a real live environment, including having	Product	Rubric	D	M	S	P	50	

		its components hosted on different machines.								

25. Absence from assessment opportunities or late submission of assessments

If a **formal test/assignment** was not written/submitted due to illness, a valid doctor's certificate needs to be presented within 5 days after the assessment was conducted. A supplementary test/assignment **may** be allowed under exceptional circumstances if proper proof was provided to the lecturer within 5 days after the assessment was conducted. The doctor's certificate must be submitted to the administrator of the department after which the student will be given proof of receipt. The office locations of the administrators are as follow:

- Soshanguve South campus: Dr Lebogang Seoketsa, building 20 room G04;
- Soshanguve South campus: Ms Molly Moche, building 10 room 103
- eMalahleni campus: Ms Angelina Bhembe, building 14 room G37;
- Polokwane campus: Ms Maesela Lebelo, building 1 room 247),

If the doctor's certificate is proven to be valid by the department, a student will be given an opportunity to write a sick test. Disciplinary measures will be instituted against any student found to have submitted an invalid doctor's certificate."

Please note that at the end of each assessment, the method of submission will be communicated. For some assessments you might be required to submit your assessment via a web interface. Failure to do so correctly will result in a zero mark. No supplementary test/assignment will be allowed if you fail to submit your assessment according to the submission instructions stated in the assessment question.

The **Sick test** will be written at the end of the semester (before the predicate mark is calculated) as is indicated in section 26. Please note that only one sick test will be allowed and that the mark for this test will replace the mark of the test that was missed.

The **supplementary** exam will be written at the end of the semester after the main exam as is indicated in section 26. Please note that the mark for this supplementary paper will replace the mark of the main exam paper.

If a student is left with one module remaining, he/she may apply for an **exit exam** but only if the student has not written a supplementary exam for this outstanding module. The exit exam will be written at the end of the semester after the supplementary exam. The dates on when this paper is to be written will be communicated by the exams department. Please note that a student may only write either the supplementary exam or the exit exam. If a student has written the supplementary exam he/she will not be allowed to write the exit exam. Please note that the mark for this exam paper will replace the mark of the exam paper that was failed.

26. Assignments

You are required to complete multiple assignments as is shown in the assessment schedule. It might be possible that a smaller number of assignments are completed if some unforeseen event arises. Please make sure that you hand in your assignment in the correct file format as specified by the assignment. Also make sure that your student number appears in the file name that is submitted to the lecturer.

When completing your assignment, you should ask yourself the following questions:

- Did I follow the technical specifications correctly?
- What skills is the lecturer trying to teach me?
- Did the work product achieve its goal?

Assignments must be completed and handed in on the due date as is specified on the schedule of this study guide. If an assignment is late, the lecturer can deduct up to 20% from the total assessment mark per day that the assignment is late. This penalty includes weekends.

If you are required to hand in a paper-based assignment you must follow the specification below:

1. All assignments must be the author's original work. No plagiarism will be tolerated! (IE: NO COPY AND PASTING)
2. Relevant cover page (as provided in this Document)
3. Typed in Calibri 11
4. 1.5 spacing, printed on one side of the page only
5. Top and Bottom margins 2.54cm (Standard margins)
6. Left and right margins 3.17cm (Standard margins)
7. A copy of all resources should be attached at the end of the bibliography
8. Full bibliography must be supplied; failure to do so will automatically fail you in the assignment. Please follow the Harvard method of referencing. If you don't know this method, ask the lecturer.
9. Unless otherwise stated stapled in the top left-hand corner.
10. Each assignment must be kept separate DO NOT bind multiple assignments together if it is to be submitted on the same day.

27. Assessment Administration

Test administration:

1. Students will write the test.
2. The Lecturer/s will mark the test.
3. The Lecture will hand back the marked scripts to each student and discuss common problems encountered. At the end of this session the marked scripts need to submitted back to the lecturer.
4. If there are any queries regarding the test, these queries must be submitted to the lecturer within 5 days after the marked scripts were discussed. NO marks changes will be made after 5 days.

Assignments administration:

1. Student completes the assignment.
2. The student must then submit the assignment as is stated in the assignment question.
3. The lecturer will mark the assignment or web test.
4. If there are any queries regarding the assessment, these queries must be submitted to the lecturer within 5 days after the marks were published. NO marks changes will be made after 5 days.

28. Mode of delivery

The mode of delivery for this module will be both contact and online. This module will be presented on Saturdays during the day.

From 5PM each day, the computer laboratories will be made available for students to practice and do homework. Student assistants are made available in these laboratories to assist students with homework related problems. These assistants are not just there to open and close the labs, please make use of their assistance.

29. Quality assurance

The quality of the study guide is maintained by the Quality Assurance committee of Tshwane University of Technology.

Students evaluate the study guide using the lecturer assessment questionnaire that is completed twice a semester.

The quality of the question papers and memorandums is maintained by an external moderator and internal moderator with qualifications higher than the year the module is being presented in.

The quality of the marking is overseen by the external moderator that ensures the marking of the scripts is fair. A moderator report is completed for each major assessment.

30. Industry related learning

Where possible guest speakers will come and address the class on advanced programming issues.

31. Plagiarism

The following is an extract from TUT's plagiarism policy. Reference: RIPPOL067.

All students have a moral obligation and responsibility to maintain the following academic integrity principles in the production and presentation of academic outputs, regardless of the presentation format and/or work type:

Each student should only submit his/her own original academic work, except when formal group work was required in the production of the academic output;

Each student should accurately indicate in all academic outputs when information is used that was produced by another scholar by referencing it in accordance with a recognised referencing convention system;

No student should use, present or submit someone else's electronic works, multimedia products or artistic works as if it is his/her own;

Each student should accurately indicate the download/access date and the uniform resource locator (URL) of the internet web page when information is used from a website, web page or other electronic source;

No student should allow another person/s to use or copy from his/her academic output and present it as their own work;

Each student is required to attach a signed Declaration of Originality (see Annexures A and B) for each academic output submission (e.g. assignment, project, manuscript, dissertation and thesis); and Each student has the responsibility to request assistance from staff members should they require guidance and/or advice about plagiarism in their academic outputs.

Students have a moral obligation to report plagiarism incidents in academic and/or research environments. All whistle-blowers are protected in terms of the Policy on Prevention of Fraud, Corruption and Theft (Policy #: VCPOL010).

Plagiarism is a form of misconduct. The relevant part (Chapter 15 – Student Discipline) of the Prospectus, Part 1 (Students' rules and regulations) read as follows:

"Any student who contravenes the provisions of rule 15.1 of the disciplinary code is guilty of misconduct and will be dealt with in terms of the disciplinary code for students …

15.1.16 Handing in any written assignment for assessment in which the essential parts of the assignment have been copied from the work of another person, or any form of plagiarism."

All students must be fully aware that plagiarism offences/penalties can seriously affect their academic status and progress at TUT and other tertiary institutions. In the most serious cases, it can result in dismissal from the University and/or formal cancellation/retraction of current/previously submitted academic outputs. In addition, the University may indicate the nature and outcome of all plagiarism offences/penalties when it is required to provide a reference or conduct statement for the particular student.

32. Glossary

A "**Module coordinator**", is a lecturer responsible for coordinating the activities within the module. A module coordinator is responsible for setting up the study guide and ensuring that all the participants stay on schedule.

A "**Secondary module coordinator**", is a lecturer responsible for assisting the module coordinator with activities that scaled due to large student number quantities within the module.

A "**Site coordinator**", is a lecturer responsible for coordinating the activities (within the module) on a learning site. The site coordinator is responsible for reporting all deviations from the activities scheduled to the module coordinator.

A "**Custodian department**", is the department responsible for managing the module and all the resources required by the module.

ADDITIONAL DOCUMENTATION

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